

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461163.7578 +/- 0.0069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.051 +/- 0.051
Transit depth (Rp/Rs)^2	0.26 +/- 0.52 [%]
Orbital Inclination (inc)	84.65 +/- 1.83
Airmass coefficient 1 (a1)	1.78 +/- 1.12
Airmass coefficient 2 (a2)	-0.43 +/- 0.34
Scatter in the residuals of the lightcurve fit is	62.92 %
Best Comparison Star	None
Optimal Aperture	1.59
Optimal Annulus	6.16
Transit Duration (day)	0.035 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.051 ± 0.051 vs 0.1326 (deviation 1.29σ)
Duration fit vs published	0.035 d vs 0.1238 d (deviation 1.75σ)
Mid-time O–C	-11.5 min
Depth SNR	0.8
Residual scatter	62.92 %

Light curve

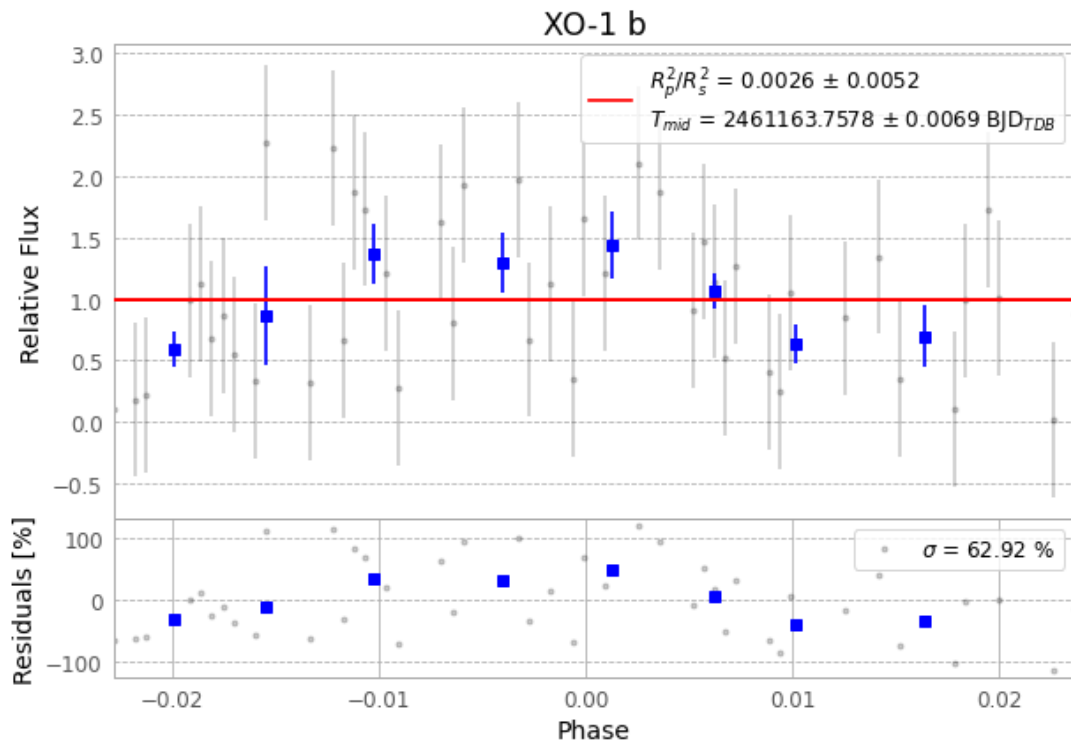


Figure 1 — FinalLightCurve_XO-1 b_2026-05-02.png (SHA-256 856c1a93fee4d35c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 268], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 343972c872d96160...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

93 FITS frames (61 MB), XO-1260503035115.FITS → XO-1260503082715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-260503.log (423f922e5010...), FinalLightCurve_XO-1 b_2026-05-02.png (856c1a93fee4...), FinalLightCurve_XO-1 b_2026-05-02.csv (c4448bc67984...)

Compute resources

Wall time 170.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.